

Dynamic programming (class 7 & 8)

1. (a) Give a $O(n^2)$ DP algorithm for finding one of the longest palindromic subsequences of $X = (x_1, x_2, \dots, x_n)$
(b) Run for CADBCDBBA.
2. Subset sum problem. Given a multiset of n of integers $(a_1, \dots, a_n)$ and a target $T \in \mathbb{N}$. Is there a $I \subseteq \{1, \dots, n\}$ such that $\sum_{j \in I} a_j = T$? Give a $O(nT)$ DP algorithm for deciding whether there it is true or not.
3. (a) Knapsack problem, given n treasures of weight $w_i \in \mathbb{N}^+$ and value $v_i \in \mathbb{N}^+$ ($1 \leq i \leq n$). Maximize your profit if your knapsack has a capacity of W . Give a $O(nW)$ DP algorithm. Is it a polynomial algorithm?
(b) Run for the instance $\{(1, 2), (3, 5), (3, 6), (4, 7), (2, 3)\}$.
4. (a) Given n exercises of complexity $m_i \in \mathbb{N}^+$ (number of minutes to solve) and score $p_i \in \mathbb{N}^+$ ($1 \leq i \leq n$). You need P points to pass (better version: to get grade 5). Give a $O(nP)$ DP algorithm to minimize the number of minutes spent.
(b) Run for the instance $\{(1, 3), (1, 2), (2, 4), (2, 6), (3, 7)\}$.
5. Give a DP algorithm to find an independent set of maximum weight in a tree if vertices are weighted (possibly with negative numbers).
6. (a) Find a triangulation of minimum weight in a convex polygon if each diagonal has a weight. Give a $O(n^3)$ DP algorithm.
(b) Run for $n = 6$ and $w(1, 3) = 5, w(1, 4) = 2, w(1, 5) = 3, w(2, 4) = 3, w(2, 5) = 4, w(2, 6) = 4, w(3, 5) = 5, w(3, 6) = 1, w(4, 6) = 2$.